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# Étale cohomology of curves

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**Lemma 0.1.** *Let  $X$  be a finite scheme over a separably closed field  $k$ . Then  $H^i(X, \mathcal{F}) = 0$  for  $i > 0$  and for all  $\mathcal{F} \in \text{Sh}(X_t)$ .*

*Proof.* Since the structure map  $f : X \rightarrow \text{Spec}(k)$  is finite,  $f_* : \text{Sh}(X_t) \rightarrow \text{Sh}(\text{Spec}(k)_t)$  is exact. Therefore, the derived functor  $R^q f_*$  vanishes for every  $q > 0$ .

Consider the Grothendieck spectral sequence

$$E_2^{p,q} = H^p(\text{Spec}(k)_t, R^q f_* \mathcal{F}) \implies H^{p+q}(X_t, \mathcal{F})$$

By the vanishing of  $R^q f_*$  we get that  $H^p(\text{Spec}(k)_t, f_* \mathcal{F}) \cong H^p(X_t, \mathcal{F})$ . Since  $k$  is separably closed, it is a strictly henselian ring, so global sections are isomorphic to stalks. Hence global sections are exact, and the cohomology vanishes.  $\square$

**Lemma 0.2.** *Let  $X$  be a scheme,  $i : Z \rightarrow X$  a closed immersion, and  $\mathcal{F} \in \text{Sh}(X_t)$  be supported on  $Z$ . Then the canonical morphism  $\eta : \mathcal{F} \rightarrow i_* i^* \mathcal{F}$  is an isomorphism.*

*Proof.* We check that  $\eta$  is an isomorphism on stalks. Let  $\bar{x}$  be a geometric point of  $X$ , and let  $x$  be the underlying point. Since  $i$  is a closed immersion, it is finite. Hence, by Stacks project [Tag 0DK3](#), we have

$$(i_* i^* \mathcal{F})_{\bar{x}} \cong \prod_{\bar{z}} (i^* \mathcal{F})_{\bar{z}},$$

where the product is taken over the geometric points  $\bar{z}$  of  $Z$  lying over  $\bar{x}$ .

If  $x \notin Z$ , this product is empty, hence

$$(i_* i^* \mathcal{F})_{\bar{x}} = 0.$$

Since  $\mathcal{F}$  is supported on  $Z$ , also  $\mathcal{F}_{\bar{x}} = 0$ , so  $\eta_{\bar{x}}$  is an isomorphism.

If  $x \in Z$ , then  $\bar{x}$  factors uniquely through  $Z$ , say as a geometric point  $\bar{z}$  of  $Z$ . Hence the product has exactly one factor, and

$$(i_* i^* \mathcal{F})_{\bar{x}} \cong (i^* \mathcal{F})_{\bar{z}} \cong \mathcal{F}_{\bar{x}}$$

$\square$

**Corollary 0.1** (of Lemma 0.2). *Let  $X$  be a scheme and  $\mathcal{F} \in X_t$  supported on finitely many closed points of  $X$ . Then*

$$\mathcal{F} \cong \bigoplus_{x \in \text{supp}(\mathcal{F})} x_* \mathcal{F}_x$$

*Proof.* We apply Lemma 0.2 with  $Z := \{x \in X : x \text{ is a geometric point}\}$  and  $i : Z \rightarrow X$ . We then have that

$$\mathcal{F} \cong i_* i^* \mathcal{F}$$

and

$$i_* i^* \mathcal{F} \cong i_* \left( \bigoplus_{x \in \text{supp}(X)} \mathcal{F}_x \right) \cong \bigoplus_{x \in \text{supp}(X)} x_* \mathcal{F}_x$$

where the latter isomorphism holds because the points are finitely many.  $\square$

**Lemma 0.3.** *Let  $X$  be a scheme,  $x$  be a geometric point of  $X$ , and  $\mathcal{F} \in \text{Sh}(x_t)$ . Then for every  $i > 0$*

$$R^i x_* \mathcal{F} = 0$$

*In particular,*

$$H^i(X_t, x_* \mathcal{F}) = 0$$

*for all  $i > 0$ .*

*Proof.* Consider an exact sequence

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$$

of étale sheaves over  $x_t$ . We have that  $x_* \mathcal{G} \rightarrow x_* \mathcal{H}$  is surjective, therefore  $R^i x_* \mathcal{F} = 0$ . As a consequence, with the same reasoning used in the proof of Lemma 0.1 using spectral sequence we obtain  $H^i(X_t, \mathcal{F}) = 0$ .  $\square$

**Lemma 0.4.** *Let  $X$  be a quasi-compact quasi-separated (qcqs) scheme, and  $\mathcal{F} \in \text{Sh}(X_t)$  be a torsion sheaf. Then  $H^i(X, \mathcal{F})$  is a torsion group for all  $i \geq 0$ .*

*Proof.* Let  $\mathcal{F}[n]$  be the kernel of the multiplication by  $n$  map  $\mathcal{F} \xrightarrow{n} \mathcal{F}$ . We have an isomorphism

$$\varinjlim (\mathcal{F}[n]) \rightarrow \mathcal{F}$$

because it is an isomorphism on stalks:

$$(\varinjlim \mathcal{F}[n])_x \cong \varinjlim ((\mathcal{F}[n])_x) \cong \varinjlim ((\mathcal{F}_x)[n]) \cong \mathcal{F}_x$$

Since  $X$  is qcqs, the cohomology commutes with colimits:

$$H^i(X_t, \mathcal{F}) \cong H^i(X_t, \varinjlim (\mathcal{F}[n])) \cong \varinjlim (H^i(X_t, \mathcal{F}[n]))$$

Since the multiplication by  $n$  map is the zero map on  $\mathcal{F}[n]$ , in cohomology it is again the zero map (being the cohomology functor additive). Therefore, the cohomology is a colimit of torsion groups, and therefore it is a torsion group.  $\square$

**Lemma 0.5.** *Let  $k$  be a separably closed field,  $K/k$  a transcendence degree 1 field extension, and  $L/K$  be an algebraic and separable field extension. Then  $L' := (L \otimes_k \bar{k})_{\text{red}}$ , where  $\bar{k}$  denotes an algebraic closure, is a field with  $C_1$  property and  $L'/L$  is purely inseparable.*

*Proof.* We first show that  $L'/L$  is purely inseparable. If  $\text{char}(k) = 0$ , then  $k$  is already algebraically closed, so  $\bar{k} = k$ , hence  $L' = L$ , and there is nothing to prove. If  $\text{char}(k) = p > 0$ , it is easy to verify that for every  $x \in L \otimes_k \bar{k}$  we get for a suitable  $n$  that  $x^{p^n} \in L$  (write  $x$  as sum of elementary tensor). Thus, passing to the reduction give the purely inseparable extension.

It remains to prove that  $L'$  is  $C_1$ . Since  $L/k(t)$  is algebraic,  $A := L \otimes_k \bar{k}$  is integral over  $k(t) \otimes_k \bar{k}$ . Now  $t$  is transcendental over  $k$ , so

$$k(t) \otimes_k \bar{k} \cong \bar{k}(t).$$

Hence  $A$  is integral over the field  $\bar{k}(t)$ , and therefore  $L'/\bar{k}(t)$  is algebraic (being  $L'$  a quotient of  $A$ ).

Now  $\bar{k}$  is algebraically closed, so by [Tsen's theorem](#) the field  $\bar{k}(t)$  is  $C_1$ . Since algebraic extensions of  $C_1$ -fields are again  $C_1$ , it follows that  $L'$  is  $C_1$ .  $\square$

**Theorem 0.1.** *Let  $C$  be an algebraic curve (of dimension 1, finite type, and separated over a field) over a separably closed field  $k$ . Then*

$$H^i(C, \mathcal{O}_{C,t}) = \begin{cases} \mathcal{O}_C(C)^\times & i = 0 \\ \text{Pic}(C) & i = 1 \\ 0 & i > 1 \text{ and } k \text{ is algebraically closed} \\ p^\infty \text{ torsion group} & i > 1 \text{ and } k \text{ is non perfect with } \text{char}(k)=p \end{cases}$$

*Proof.* The case  $i = 0$  is clear. We will not prove the case  $i = 1$ , since it follows from the interpretation of first cohomology in terms of torsors. More precisely,  $H^1(C_{\text{ét}}, \mathcal{O}_{C,\text{ét}}^\times)$  classifies  $\mathcal{O}_{C,\text{ét}}^\times$ -torsors on  $C_{\text{ét}}$ , see [Tag 03AJ](#). These torsors are equivalent to line bundles on  $C$ , thus we obtain

$$H^1(C_{\text{ét}}, \mathcal{O}_{C,\text{ét}}^\times) \cong \text{Pic}(C),$$

see [Tag 040E](#) for the proof.

Let's study the case of  $i > 1$ .

- **Step 1:** suppose the curve to be regular and connected.

In particular, the curve is integral, since every stalk is a noetherian regular local ring (hence reduced) and since every point belongs to exactly one irreducible component. Since irreducible components are closed (because  $C$  is a noetherian topological space), the curve is indeed integral.

Let  $\eta$  be the generic point. Consider the following exact sequence:

$$0 \rightarrow \mathcal{O}_{C,\text{ét}}^\times \rightarrow \eta_* \mathcal{O}_{\eta,\text{ét}}^\times \rightarrow \bigoplus_{c \in C \text{ closed}} c_* \mathbb{Z} \rightarrow 0 \quad (1)$$

where the second map is the divisor map. This sequence is exact since on the stalk at a geometric point  $\bar{x}$  over a closed point it is given by

$$0 \rightarrow (\mathcal{O}_{C,\bar{x}}^{\text{sh}})^\times \rightarrow (\text{Frac}(\mathcal{O}_{C,\bar{x}}^{\text{sh}}))^\times \rightarrow \mathbb{Z} \rightarrow 0$$

and the right-hand side map is the valuation map.

Taking cohomology of sequence 1 gives that

$$H^i(C, \mathcal{O}_{C,\text{ét}}^\times) \cong H^i(C, \eta_* \mathcal{O}_{\eta,\text{ét}}^\times) \quad (2)$$

for all  $i > 1$ . Indeed, the sheaf  $\bigoplus_{c \in C \text{ closed}} c_* \mathbb{Z}$  is the filtered colimit of the finite sums  $\bigoplus_{c \in S} c_* \mathbb{Z}$ , where  $S$  ranges over the finite subsets of closed points of  $C$ . By Corollary 0.1, each such finite sum has vanishing higher cohomology. Since  $C$  is qcqs, filtered colimits of abelian sheaves on  $C_{\text{ét}}$  commute with cohomology by the Stacks Project Tag 073D. Therefore

$$H^i \left( C_{\text{ét}}, \bigoplus_{c \in C \text{ closed}} c_* \mathbb{Z} \right) = 0$$

for every  $i > 0$ , and the claimed isomorphism follows from the long exact sequence in cohomology.

By Tag 03Q9, we have for every geometric point  $c$  of  $C$

$$(\mathbf{R}^q \eta_* \mathcal{O}_{\eta,\text{ét}}^\times)_c \cong H^q((\eta \times_C \text{Spec}(\mathcal{O}_{C,c}^{\text{sh}}))_t, (\mathcal{O}_{\eta,\text{ét}}^\times)_{|\eta \times_C \text{Spec}(\mathcal{O}_{C,c}^{\text{sh}})}) \quad (3)$$

As  $\eta \times_C \text{Spec}(\mathcal{O}_{C,c}^{\text{sh}}) = \text{Spec}(\text{Frac}(\mathcal{O}_{C,c}^{\text{sh}}))$ , write  $L$  for  $\text{Frac}(\mathcal{O}_{C,c}^{\text{sh}})$ , so that the restriction of  $\mathcal{O}_{\eta,\text{ét}}^\times$  appearing in the right-hand side of 3 is just  $\mathcal{O}_{\text{Spec}(L),\text{ét}}^\times$ . Observe that  $L$  is an algebraic separable extension of the function field  $k(C)$ . Indeed,  $\mathcal{O}_{C,c}^{\text{sh}}$  is the filtered colimit of local étale  $\mathcal{O}_{C,c}$ -algebras  $B$ , and for each such  $B$  the field  $\text{Frac}(B) = B \otimes_{\mathcal{O}_{C,c}} k(C)$  is a finite separable extension of  $k(C)$ , thus every element of  $L$  lies in some finite separable extension of  $k(C)$ .

By the relation between Galois and étale cohomology, we have

$$H^q(\text{Spec}(L), \mathcal{O}_{\text{Spec}(L),\text{ét}}^\times) \cong H^q(\text{Gal}(L^{\text{sep}}/L), (L^{\text{sep}})^\times) \quad (4)$$

By Hilbert 90, this vanishes for  $q = 1$ .

**Claim.** The group

$$H^q(\mathrm{Gal}(L^{\mathrm{sep}}/L), (L^{\mathrm{sep}})^{\times})$$

is a  $p^{\infty}$ -torsion group for every  $q \geq 2$ .

*Proof.* By Lemma 0.5,  $L' := (L \otimes_k \bar{k})_{\mathrm{red}}$  is a  $C_1$ -field with  $L'/L$  purely inseparable. It is a well-known Galois-cohomology fact (Stacks project Tag 03R8 and Tag 03RC) that for a  $C_1$ -field

$$H^i(\mathrm{Gal}(L'^{\mathrm{sep}}/L'), (L'^{\mathrm{sep}})^{\times}) = 0$$

for every  $i > 0$ . Therefore, by pure inseparability of  $L'/L$ , we get, by the isomorphism on Galois groups,

$$H^i(\mathrm{Gal}(L^{\mathrm{sep}}/L), (L^{\mathrm{sep}})^{\times}) \cong H^i(\mathrm{Gal}(L'^{\mathrm{sep}}/L'), (L'^{\mathrm{sep}})^{\times}) = 0$$

for every  $i > 0$ .

Consider the short exact sequence

$$0 \rightarrow (L^{\mathrm{sep}})^{\times} \rightarrow (L'^{\mathrm{sep}})^{\times} \rightarrow Q \rightarrow 0$$

where  $Q$  is the cokernel of the first map. By pure inseparability of  $L'^{\mathrm{sep}}/(L^{\mathrm{sep}})$ , we have that  $Q$  is a  $p^{\infty}$ -torsion group. Passing to cohomology, we get

$$H^{i+1}(\mathrm{Gal}(L^{\mathrm{sep}}/L), (L^{\mathrm{sep}})^{\times}) \cong H^i(\mathrm{Gal}(L^{\mathrm{sep}}/L), Q)$$

for every  $i > 0$ . Therefore, by Lemma 0.4, we get the claim.  $\square$

By isomorphisms 3, 4, and Claim, we get that

$$(R^q \eta_* \mathcal{O}_{\eta, \acute{\mathrm{e}}\mathrm{t}}^{\times})_c$$

is a  $p^{\infty}$ -torsion group at every geometric point  $c$ , thus the sheaf  $R^q \eta_* \mathcal{O}_{\eta, \acute{\mathrm{e}}\mathrm{t}}^{\times}$  is a  $p^{\infty}$ -torsion sheaf.

Consider the Grothendieck spectral sequence

$$E_2^{i,j} := H^i(C_{\acute{\mathrm{e}}\mathrm{t}}, R^j \eta_* \mathcal{O}_{\eta, \acute{\mathrm{e}}\mathrm{t}}^{\times}) \Rightarrow H^{i+j}(\eta_{\acute{\mathrm{e}}\mathrm{t}}, \mathcal{O}_{\eta, \acute{\mathrm{e}}\mathrm{t}}^{\times}).$$

Since this is a first-quadrant spectral sequence, we also get

$$E_2^{i,j} \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}] \Rightarrow H^{i+j}(\eta_{\acute{\mathrm{e}}\mathrm{t}}, \mathcal{O}_{\eta, \acute{\mathrm{e}}\mathrm{t}}^{\times}) \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}].$$

For every  $j \geq 1$ , we proved that  $R^j \eta_* \mathcal{O}_{\eta, \acute{\mathrm{e}}\mathrm{t}}^{\times}$  is a  $p^{\infty}$ -torsion sheaf. Hence  $E_2^{i,j} = H^i(C_{\acute{\mathrm{e}}\mathrm{t}}, R^j \eta_* \mathcal{O}_{\eta, \acute{\mathrm{e}}\mathrm{t}}^{\times})$  is a  $p^{\infty}$ -torsion group for every  $i \geq 0$  and every  $j \geq 1$ , by Lemma 0.4. Therefore

$$E_2^{i,j} \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}] = 0 \quad (j \geq 1).$$

Thus the localized spectral sequence has only the row  $j = 0$  possibly nonzero, and therefore it degenerates at the  $E_2$ -page. Hence for every  $n > 0$  one gets an isomorphism

$$E_2^{n,0} \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}] \cong H^n(\eta_{\text{ét}}, \mathcal{O}_{\eta, \text{ét}}^\times) \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}].$$

Now  $H^1(\eta_{\text{ét}}, \mathcal{O}_{\eta, \text{ét}}^\times) = 0$  by Hilbert-90, and for every  $n \geq 2$  the group  $H^n(\eta_{\text{ét}}, \mathcal{O}_{\eta, \text{ét}}^\times)$  is a  $p^\infty$ -torsion group by the same Galois-cohomological argument as above, applied to the field  $k(C)$ . Therefore

$$H^n(\eta_{\text{ét}}, \mathcal{O}_{\eta, \text{ét}}^\times) \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}] = 0 \quad (n > 0).$$

It follows that

$$E_2^{n,0} \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}] = 0 \quad (n > 1).$$

Since  $E_2^{n,0} = H^n(C_{\text{ét}}, \eta_* \mathcal{O}_{\eta, \text{ét}}^\times)$ , we obtain

$$H^n(C_{\text{ét}}, \eta_* \mathcal{O}_{\eta, \text{ét}}^\times) \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}] = 0 \quad (n > 1).$$

By the isomorphism 2, this gives the claim in the case of a regular connected curve.

- **Step 2:** suppose  $C$  is regular.

Apply **Step 1** to every connected component.

- **Step 3:** suppose  $C$  is reduced.

Consider the normalization  $\nu : C' \rightarrow C$  of the curve. We first prove that the natural morphism

$$\mathcal{O}_{C, \text{ét}}^\times \rightarrow \nu_* \mathcal{O}_{C', \text{ét}}^\times$$

is injective.

Let  $C_{\text{reg}}$  be the regular locus of  $C$ . It is Zariski open in  $C$ , since  $C$  is locally of finite type over a field. Moreover, since  $C$  is reduced, each generic point of  $C$  is regular, thus  $C_{\text{reg}}$  is dense. Since regular points are normal, the normalization is an isomorphism over  $C_{\text{reg}}$ . Hence

$$(\mathcal{O}_{C, \text{ét}}^\times)|_{C_{\text{reg}}} \rightarrow (\nu_* \mathcal{O}_{C', \text{ét}}^\times)|_{C_{\text{reg}}}$$

is an isomorphism.

Now let  $V \rightarrow C$  be an affine étale morphism. Since étale morphisms are open,

$$V_{\text{reg}} := V \times_C C_{\text{reg}}$$

is a dense open subset of  $V$ . After base change to  $V$ , the normalization map is still an isomorphism over  $V_{\text{reg}}$ . Therefore the map

$$\Gamma(V, \mathcal{O}_V^\times) \rightarrow \Gamma(V \times_C C', \mathcal{O}_{V \times_C C'}^\times)$$

becomes an isomorphism after restricting to the dense open subset  $V_{\text{reg}}$ .

Since  $C$  is reduced and  $V \rightarrow C$  is étale, also  $V$  is reduced. Hence restriction to a dense open subset is injective, thus

$$\Gamma(V, \mathcal{O}_V^\times) \rightarrow \Gamma(V \times_C C', \mathcal{O}_{V \times_C C'}^\times)$$

is injective. Since this holds for every affine étale  $V \rightarrow C$ , the morphism

$$\mathcal{O}_{C, \text{ét}}^\times \rightarrow \nu_* \mathcal{O}_{C', \text{ét}}^\times$$

is injective.

Now define  $Q$  as the cokernel of this injective morphism. We have an exact sequence

$$0 \rightarrow \mathcal{O}_{C, \text{ét}}^\times \rightarrow \nu_* \mathcal{O}_{C', \text{ét}}^\times \rightarrow Q \rightarrow 0 \quad (5)$$

Since the normalization is an isomorphism over  $C_{\text{reg}}$ , the sheaf  $Q$  vanishes on  $C_{\text{reg}}$ . Hence  $Q$  is supported on finitely many points of  $C$ . Therefore, by Corollary 0.1, we get

$$H^i(C_{\text{ét}}, \mathcal{O}_{C, \text{ét}}^\times) \cong H^i(C_{\text{ét}}, \nu_* \mathcal{O}_{C', \text{ét}}^\times)$$

for every  $i \geq 2$ .

Moreover, by the Stacks Project [Tag 035S](#) and [Tag 035B](#),  $\nu$  is finite. Hence, by the exactness of the direct image of a finite morphism on étale sheaves,  $\nu_*$  is exact, giving

$$H^i(C_{\text{ét}}, \nu_* \mathcal{O}_{C', \text{ét}}^\times) \cong H^i(C', \mathcal{O}_{C', \text{ét}}^\times)$$

Since  $C'$  is normal of dimension 1, it is regular, so we can apply **Step 2**.

- **Step 4:** no additional hypotheses on  $C$ .

Consider  $\mathcal{N} := \text{Nil}(\mathcal{O}_C)$ . We proceed by induction on  $n$  such that  $\mathcal{N}^{2^n} = 0$ . The case  $n = 0$  is given by **Step 3**. Assume now that  $n > 0$  and that the result is known whenever the nilradical is killed by  $2^{n-1}$ . Define  $J := \mathcal{N}^{2^{n-1}}$ , let  $C_0$  be the closed subscheme defined by it, and let  $\pi : C_0 \rightarrow C$ . Since  $J^2 = \mathcal{N}^{2^n} = 0$ , we get an exact sequence

$$0 \rightarrow J \rightarrow \mathcal{O}_{C, \text{ét}}^\times \rightarrow \pi_* \mathcal{O}_{C_0, \text{ét}}^\times \rightarrow 1$$

where the left map sends an element  $a$  to  $1 + a$  on sections. Since  $J$  is quasi-coherent,

$$H^i(C_{\text{ét}}, J) \cong H^i(C_{\text{Zar}}, J)$$

for every  $i \geq 0$ . Thus, by dimension vanishing, we get an isomorphism

$$H^i(C_{\text{ét}}, \mathcal{O}_{C, \text{ét}}^\times) \cong H^i(C_{\text{ét}}, \pi_* \mathcal{O}_{C_0, \text{ét}}^\times)$$



for every  $i > 1$ .

Since  $\pi$  is a closed immersion, it is finite, thus  $\pi_*$  is exact, giving

$$H^i(C_{\text{ét}}, \pi_* \mathcal{O}_{C_0, \text{ét}}^\times) \cong H^i((C_0)_{\text{ét}}, \mathcal{O}_{C_0, \text{ét}}^\times)$$

for every  $i \geq 0$ . Moreover, the nilradical of  $\mathcal{O}_{C_0}$  is killed by  $2^{n-1}$ . Applying the induction hypothesis, we get the claim. □

**Theorem 0.2.** *Let  $C$  a proper smooth connected curve of genus  $g$  over a separably closed field  $k$ . Let  $n \in \mathbb{N}$  coprime with  $\text{char}(k)$ . Then*

$$H^i(C_{\text{ét}}, \mu_n) = \begin{cases} \mu_n(k) & i = 0 \\ \text{Pic}^0(C)[n] & i = 1 \\ \mathbb{Z}/n\mathbb{Z} & i = 2 \\ 0 & i > 2 \end{cases}$$